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Estimating a time-dependent concordance index for survival prediction models with covariate dependent censoring

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Given a predictive marker and a time-to-event response variable, the proportion of concordant pairs in a data set is called concordance index. A specifically useful marker is the risk predicted by a survival regression model. This article extends the existing methodology for applications where the length of the follow-up period depends on the predictor variables. A class of inverse probability of censoring weighted estimators is discussed in which the estimates rely on a working model for the conditional censoring distribution. The estimators are consistent for a truncated concordance index if the working model is correctly specified and if the probability of being uncensored at the truncation time is positive. In this framework, all kinds of prediction models can be assessed, and time trends in the discrimination ability of a model can be captured by varying the truncation time point. For illustration, we re-analyze a study on risk prediction for prostate cancer patients. The effects of misspecification of the censoring model are studied in simulated data. Copyright © 2012 John Wiley & Sons, Ltd.

Keywords: censored data; concordance index; discrimination; inverse probability of censoring weighting; prediction models; survival analysis

1. Introduction

The concordance index [1] is a well-recognized measure of discrimination for models that predict a time to event. It has been reported in the medical literature more often than any other prediction model metric in the survival setting. The concordance index also plays an important role among the recommended validation metrics for biomarker discovery and the development of prediction models [2, 3].

For uncensored data, no further modeling is required, and the concordance index is very easy to interpret as the relative frequency of concordant pairs among all pairs of patients. A pair is called concordant if the individual with the shorter survival time received the higher predicted risk. For right-censored event times, the order of a pair remains unknown if the patient with the shorter follow-up time did not experience the event. Such pairs are called unusable. Ignoring unusable pairs leads to a simple statistic that Brown *et al.* discovered [4] and Harrell *et al.* made popular [1, 5]. As Brown *et al.* readily pointed out [4], the simple statistic does not use all the available information, and it estimates a population parameter that depends on the censoring distribution. This is an undesirable property.

Several methods have been proposed to consistently estimate the concordance index based on right-censored data [4, 6–9]. Notably the imputation method of Brown *et al.* [4] and the nonparametric estimates of Liu and Jin [8] and Uno *et al.* [9] do not assume a correctly specified survival regression model and are therefore generally preferred. However, both methods are not well adapted to situations where the censoring distribution depends on the predictor variables.

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In this article, we propose a class of inverse of the probability of censoring weighted (IPCW) estimators for estimating a time-truncated concordance index denoted $\mathcal{C}(t)$. The time-truncated parameter is preferred for right-censored data [7]. The proposed estimators rely on a working model for the conditional censoring distribution-given predictor variables and are based on general theory for censored and missing data [10, 11]. As shown in Section 3.2, the IPCW estimate is consistent under all of the three assumptions: (A1) conditionally independent censoring, (A2) a working model for the conditional censoring distribution that is correctly specified, and (A3) the probability of being uncensored by time t is bounded away from 0. A special working model is to assume that the censoring does not depend on the predictor variables. Therefore, the estimates of Liu and Jin [8] and Uno *et al.* [9] appear as special cases of our framework. However, a bias is incurred if the censoring distribution depends on the predictor variables as illustrated in simulated data in Section 3.3.

For the purpose of illustration, we re-analyze the data of a study on prediction models for prostate cancer patients [12]. A peculiarity of the study was that the treatment dose increased with calendar time. Patients enrolled late received higher doses, and therefore the censoring times depended heavily on the predictor variables. In Section 4, we apply the proposed IPCW estimates with a working Cox regression model for the censoring hazard function. A sensitivity study is conducted to test the results against misspecification of the working model (Section 4.6). An interesting side result is that a time-truncated version of the aforementioned simple statistic, which ignores unusable pairs, performs reasonably well in the sensitivity analysis.

2. The risk discrimination index

2.1. Setup and notation

For patient i let T_i be the time of death or other event and $X_i = (X_{i1}, \ldots, X_{iL})$ a vector of time-independent predictor variables. Observed are $\tilde{T}_i = \min(T_i, C_i)$, $\Delta_i = 1\{T_i \leqslant C_i\}$, and X_i, where C_i is the individual censoring time determined by the end of the event-free follow-up period for subject i. Denote $D_n^{\text{train}} = \left\{\tilde{T}_i, \Delta_i, X_i), i \in \mathcal{I}_n^{\text{train}}\right\}$ for a training data set in which the prediction models are derived and $D_m^{\text{test}} = \left\{\tilde{T}_i, \Delta_i, X_i), i \in \mathcal{I}_m^{\text{test}}\right\}$ for an independent validation data set in which the concordance index is estimated. We illustrate how to proceed when there is only one data set available for training and validation in Section 4.4.

Consider a risk prediction model M_n that was obtained on the basis of the data D_n^{train}. The model predicts an event probability $M_n(t, X_i)$ for patients $i \in \mathcal{I}_m^{\text{test}}$ at a suitably selected prediction horizon time t. For the definition and estimation of the concordance index, the only requirement is that the model can order any two subjects based on the information available at baseline. If $M_n(t, X_i) > M_n(t, X_j)$, then the model predicts a higher event probability (lower survival chance) until time t for person i than for person j. For the following theoretical results to hold, the prediction horizon t should be such that $P(C_i > t|X_i) > \epsilon > 0$. In practice, it should be chosen smaller than the maximum follow-up time in the training set.

2.2. Risk prediction models

We consider prediction models for medical practice and refer to [13] for applications, examples, and steps of modeling. Even though our setup does not require this, in many applications the risk prediction model will be derived from a suitably specified survival regression model. For example, a Cox regression model specifies the event probability as

$$M_n^{\text{COX}}(t, X_i) = 1 - \exp\left(-\hat{\Lambda}_0(t) \exp\left\{\hat{\beta}^T X_i\right\}\right).$$

The estimates $\hat{\Lambda}_0(t)$ of the baseline hazard rate and $\hat{\beta}$ of the L-dimensional vector of regression coefficients are obtained from the training data D_n^{train}. They do not depend on the test set. This is a proportional hazards model, where a ranking of individuals $i, j \in \mathcal{I}_m^{test}$ from the test set, viz. $M_n^{\text{COX}}(t, X_i) < M_n^{\text{COX}}(t, X_i)$, does not depend on the choice of the prediction horizon t. That is, the ranking is preserved and equivalent to $M_n^{\text{COX}}(s, X_i) < M_n^{\text{COX}}(s, X_i)$ at all time horizons s at which $\hat{\Lambda}_0(s)$ is defined. An extension of this model is a stratified Cox regression model in which the baseline hazard is allowed to depend on the covariates. Then the ranking of two test set individuals from different strata depends on the prediction horizon. The nonparametric additive Aalen model [14] features

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time-varying regression coefficients also for continuous covariates. For an L-dimensional vector of time-varying coefficients η and corresponding estimates $\hat{\eta}$, the Aalen model estimates the event risk as

$$M_n^{\text{Add}}(t, X_i) = 1 - \exp\left\{-\hat{\eta}^T(t) X_i\right\}.$$

Because the regression coefficients $\eta(t)$ are time varying, the ranking of two test set individuals obtained by the additive Aalen model generally depends on the prediction horizon t.

A random survival forest [15] is an example of a model that does not fit into the classical framework of regression models. It is an ensemble of bootstrap survival trees that predict event probabilities according to the following formula:

$$M_n^{\text{RSF}}(t, X_i) = 1 - \exp\left(-\frac{1}{B}\sum_{b=1}^{B} \hat{H}_b(t|X_i)\right).$$

Here B is the number of bootstrap samples drawn from the training set and H_b is the Nelson–Aalen estimate of the hazard function obtained with the patients in the training data set who share the terminal node of the survival tree grown in the bth bootstrap sample with X_i.

In Section 4, we apply these survival regression strategies to derive prediction models for the outcome of prostate cancer patients.

2.3. *The unrestricted concordance index*

We fix a prediction horizon t. The unrestricted concordance index is defined as

$$\tilde{C}(t) = E_{ij}\left\{1\{M_n(t, X_i) > M_n(t, X_j)\} \mid T_i < T_j, D_n^{\text{train}}\right\}. \tag{1}$$

Here and in what follows E_{ij} denotes that the expectation is taken with respect to the data of two subjects $i, j \notin \mathcal{I}_n^{\text{train}}$ that were not included in the training data D_n^{train}. If the event time of patient i is smaller than the event time of patient j then a good model should have predicted the higher survival probability for subject j. Thus, the unrestricted concordance index is the ability of the model to rank the event times based on the predicted risks. This makes most sense when the predicted order of two subjects does not depend on time, that is, when $M_n(t, X_i) > M_n(t, X_j)$ implies that $M_n(s, X_i) > M_n(s, X_j)$ for all time points s. The truncated concordance index defined subsequently can also be used to assess models in which the risk order of two patients changes over time.

The parameter $\tilde{C}(t)$ defined in (1) has a nice and useful interpretation, but it is not generally identifiable in a medical study with limited follow-up. The reason is that it is not possible to assess the order of the event times that occur beyond the maximum follow-up time of the study. For the same reason, it is not possible to estimate the (unrestricted) expected survival time.

2.4. *Truncated concordance and discrimination curve*

As in [7], consider the following truncated version of the concordance index:

$$C_t(t) = E_{ij}\left\{1\{M_n(t, X_i) > M_n(t, X_j)\} \mid T_i < T_j, T_i \leqslant t, D_n^{\text{train}}\right\}, \tag{2}$$

where again the expectation is taken with respect to the data of two subjects $i, j \notin \mathcal{I}_n^{\text{train}}$ and where the subscript t refers to the truncation time. Hence, $\tilde{C}(t) = C_\infty(t)$. It is possible to define $C_s(t)$ with prediction horizon t different from the truncation time s. However, for simplicity of presentation, we will only consider situations where the concordance index is truncated at the same time where also the event risks are predicted and we will write $C_t(t) = C(t)$ in what follows.

The truncated concordance index $C(t)$ is interpreted as the ability of the model to rank the event times that occur before time t. By varying the prediction horizon, we obtain a *risk discrimination curve* $t \mapsto C(t)$ that can be identified from right-censored data until t reaches a certain maximum time point. The maximum time point is similar to the maximum point of definition of the Kaplan–Meier estimate. For proportional hazard models, the discrimination curve will only vary because of different truncation times. For more flexible models, such as stratified Cox regression models or Aalen regression models, the discrimination curve will also vary because of changes of the ranking of subjects over time.

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3. Estimation

3.1. Ignoring unusable pairs

The simple concordance statistic was derived from a modification of Kendall's rank correlation for censored data [1, 4, 5]. In right-censored data, it ignores pairs of patients whenever the earlier event time was censored ($\Delta_i = 0$). With notation $\widetilde{N}_i(t) = 1\left\{\widetilde{T}_i \leqslant t, \Delta_i = 1\right\}$, we define a truncated version of the simple statistic as

$$\hat{C}_{\text{simp}}(t) = \frac{\frac{1}{m^2}\sum_{i=1}^{m}\sum_{j=1}^{m} 1\{M_n(t, X_i) > M_n(t, X_j)\}1\left\{\widetilde{T}_i < \widetilde{T}_j\right\}\widetilde{N}_i(t)}{\frac{1}{m^2}\sum_{i=1}^{m}\sum_{j=1}^{m} 1\left\{\widetilde{T}_i < \widetilde{T}_j\right\}\widetilde{N}_i(t)} = \frac{\hat{A}_{\text{simp}}(t)}{\hat{B}_{\text{simp}}(t)}. \tag{3}$$

The statistic can be interpreted as the *frequency of definitely concordant pairs out of definitely concordant or definitely discordant pairs*, because if $1\left\{\widetilde{T}_i < \widetilde{T}_j\right\} = 1$ and $\Delta_i = 1$, then it can be deduced from the observable data that $T_j > T_i$. The advantage of the simple statistic is that it does not require further modeling. The main disadvantage of the simple statistic is that it has a censoring bias when considered an estimate of the population parameter (2), see Appendix A.

3.2. IPCW estimate

The truncated concordance index (2) can be expressed as a functional of the conditional survival distribution $S(t \mid x) = P(T_i > t \mid x)$:

$$\begin{aligned} C(t) &= \frac{E_{ij}\left\{1\{M_n(t, X_i) > M_n(t, X_j)\}1\{T_i < T_j, T_i \leqslant t\}|D_n^{\text{train}}\right\}}{P(T_i < T_j, T_i \leqslant t)} \\ &= \frac{E_{ij}\left\{1\{M_n(t, X_i) > M_n(t, X_j)\}E_{ij}\{1\{T_i < T_j, T_i \leqslant t\}|X_i, X_j\}|D_n^{\text{train}}\right\}}{P(T_i < T_j, T_i \leqslant t)} \\ &= \frac{E_{ij}\left\{1\{M_n(t, X_i) > M_n(t, X_j)\}\int_0^t S(u \mid X_j)S(\mathrm{d}u \mid X_i)|D_n^{\text{train}}\right\}}{E_{ij}\left\{\int_0^t S(u \mid X_j)S(\mathrm{d}u \mid X_i)\right\}}. \end{aligned} \tag{4}$$

The first step in the previous display follows because $M_n(t, X_i)$ is a function of X_i that does not depend on T_i and therefore can be extracted from the conditional expectation given X_i, X_j. For the second step, note that $E_{ij}\{1\{T_i < T_j\}|X_i, X_j\} = P(T_i < T_j|X_i, X_j) = \int_0^\infty P(T_j > u|X_j)S(\mathrm{d}u \mid X_i) = \int_0^\infty S(u|X_j)$ $S(\mathrm{d}u \mid X_i)$.

The IPCW estimates of the concordance index (described subsequently) are obtained by substituting IPCW estimates for $S(t|x)$. See [10] for general theory and [16] for a similar application. Denote $G(t \mid x) = P(C_i > t \mid x)$ for the conditional survival function of the censoring times. We assume throughout that

$$C_i \perp\!\!\!\perp T_i \mid X_i \quad \text{(conditional independence)}. \tag{A1}$$

Under (A1) we have $P\left(\widetilde{T}_i > t|x\right) = S(t \mid x)G(t \mid x)$ and

$$P(\Delta_i = 1 \mid X_i = x, T_i = t) = P(C_i \geqslant t \mid X_i = x) = G(t- \mid x). \tag{5}$$

We also assume that the probability of being uncensored at the prediction horizon is bounded away from 0:

$$G(t \mid x) > \epsilon > 0 \text{ uniformly in } x. \tag{A2}$$

We now describe a class of IPCW estimates for the truncated concordance index. $\mathcal{G}$ denotes a working model for G and we assume an estimate $\widehat{G}_m$ that converges in probability to $G^* \in \mathcal{G}$. If the working model happens to be correctly specified, then $G \in \mathcal{G}$ and $\widehat{G}_m$ are consistent:

$$\widehat{G}_m \overset{m\to\infty}{\to} G^* = G. \tag{A3}$$

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If the working model is misspecified, then $\widehat{G}_m \overset{m\to\infty}{\to} G^* \neq G$. With weights $\widehat{W}_{ij} = \widehat{G}_m\left(\widetilde{T}_i \mid X_j\right)\widehat{G}_m\left(\widetilde{T}_i- \mid X_i\right)$, we define the IPCW estimate on the basis of the validation data as

$$\hat{\mathcal{C}}_{\text{ipcw}}(t) = \frac{\frac{1}{m^2}\sum_{i=1}^m\sum_{j=1}^m 1\{M_n(t,X_i) > M_n(t,X_j)\}1\left\{\widetilde{T}_i < \widetilde{T}_j\right\}\widetilde{N}_i(t)\widehat{W}_{ij}}{\frac{1}{m^2}\sum_{i=1}^m\sum_{j=1}^m 1\left\{\widetilde{T}_i < \widetilde{T}_j\right\}\widetilde{N}_i(t)\widehat{W}_{ij}}^{-1} = \frac{\hat{A}_{\text{ipcw}}(t)}{\hat{B}_{\text{ipcw}}(t)}. \quad (6)$$

The estimator is well defined at time t if $\widehat{G}_m(t \mid x) > 0$ for all x. We show in Appendix B that $\hat{\mathcal{C}}_{\text{ipcw}}(t)$ consistently estimates $\mathcal{C}(t)$ with increasing size of the independent validation sample and under assumptions (A1)–(A3). In applications with ties between censored times and event times, the distinction between $\widehat{G}_m\left(\widetilde{T}_i \mid X_i\right)$ and its left-hand limit $\widehat{G}_m\left(\widetilde{T}_i- \mid X_i\right)$ is important as it may stabilize the weights.

3.3. Simulation results

In this section, we illustrate the potential bias incurred using a misspecified working model for the censoring distribution. For this purpose, a simple simulation setting with a single binary predictor is sufficient. For a simulation study of a more realistic situation, see Section 4.6.

We repeated the following steps for different sample sizes: $N = 100, 200, 500, 1000$. We simulated the binary marker on the basis of a Bernoulli distribution with success probability 60%. We generated event times on the basis of a Weibull regression model with shape parameter equal to 2, scale parameter equal to 0.01 [17], and repeated for different effects of the binary marker $\beta = (0, 0.7, 1.3)$. We generated censoring times on the basis of a Weibull regression model with shape parameter equal to 1, scale parameter equal to 0.1, and repeated for different effects of the binary marker $\gamma = (0, 0.7, 1.3)$. This yielded censoring levels between 40% and 80%. For each (β, γ) constellation, we chose a prediction horizon t_0 such that the expected number at risk at t_0 was roughly 10% of the data. We held this time point fixed for all sample sizes. We removed the results of a few simulation runs, where the absolute number at risk at time t_0 was below 5. We derived predictions from a Cox regression model that was fitted to a fixed training data set of size N. The true value of $\mathcal{C}(t_0)$ for this model was approximated by the average of 100 estimates of $\mathcal{C}(t_0)$ obtained in large *uncensored* validation data sets ($N = 30{,}000$). For each (β, γ, N) constellation, we simulated 10,000 validation data sets simulated independently of the training data. In each set, we computed the truncated version of the simple statistic $\left(\hat{\mathcal{C}}_{\text{simp}}\right)$ and three different IPCW estimates based on three different working models: (1) independent censoring, with the weights estimated using the marginal Kaplan–Meier estimate for the censored times $\left(\hat{\mathcal{C}}_{\text{ipcw.1}}\right)$; (2) working Cox regression model for the censored times $\left(\hat{\mathcal{C}}_{\text{ipcw.2}}\right)$; and (3) stratified Kaplan–Meier estimate for the censored times $\left(\hat{\mathcal{C}}_{\text{ipcw.3}}\right)$. We computed for $M = 10{,}000$ the mean squared error $\text{MSE} = 1/M * \sum_{i=1}^M \left\{\mathcal{C}(t_0) - \hat{\mathcal{C}}(t_0)\right\}^2$ and the absolute bias $= \mid \mathcal{C}(t_0) - 1/M \sum_{i=1}^M \hat{\mathcal{C}}(t_0) \mid$. Note that model (1) is misspecified whenever $\gamma > 0$ and that in all settings the models (2) and (3) were correctly specified.

Common in all settings where the binary marker has an effect on survival ($\beta > 0$), the simulation results show smaller MSE for the IPCW estimates $\hat{\mathcal{C}}_{\text{ipcw.2}}$ and $\hat{\mathcal{C}}_{\text{ipcw.3}}$ as compared with the truncated version of the simple statistic. Table I shows the results for sample size 200. The table also shows that the bias of $\hat{\mathcal{C}}_{\text{simp}}$ and $\hat{\mathcal{C}}_{\text{ipcw.1}}$ increases with increasing γ. This was the case for all other sample sizes (results not shown). For sample sizes 100 (results not shown) and 200, if the effect on the censoring is stronger than the effect on survival ($\gamma > \beta$), the bias of $\hat{\mathcal{C}}_{\text{ipcw.2}}$ and of $\hat{\mathcal{C}}_{\text{ipcw.3}}$ was larger than the bias of the simple estimate. However, for sample sizes 500 and 1000, this phenomenon was only seen for $\beta = 0$ (results not shown).

Figure 1 shows boxplots over 10,000 simulations for all sample sizes for $\beta = 1.3$ and $\gamma = 1.3$. The figure clearly illustrates the large-sample bias of the estimates $\hat{\mathcal{C}}_{\text{simp}}$ and $\hat{\mathcal{C}}_{\text{ipcw.1}}$, consistency of $\hat{\mathcal{C}}_{\text{ipcw.2}}$ and $\hat{\mathcal{C}}_{\text{ipcw.3}}$, and uniformly less variability of the latter.

4. The prostate cancer study

The main objective of the study published by Kattan *et al.* [12] was to develop a nomogram that accurately predicts the response to radiotherapy treatment of prostate cancer patients. Various statistical

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Table I. Results of 10,000 simulations with a single binary predictor for sample size 200 and varying effects of the predictor on the survival distribution (β) and the censoring distribution (γ).

$N = 200$				MSE				Absolute bias			
β	γ	$\mathcal{C}(t_0)$	Cens	$\hat{\mathcal{C}}_{\text{simp}}$	$\hat{\mathcal{C}}_{\text{ipcw.1}}$	$\hat{\mathcal{C}}_{\text{ipcw.2}}$	$\hat{\mathcal{C}}_{\text{ipcw.3}}$	$\hat{\mathcal{C}}_{\text{simp}}$	$\hat{\mathcal{C}}_{\text{ipcw.1}}$	$\hat{\mathcal{C}}_{\text{ipcw.2}}$	$\hat{\mathcal{C}}_{\text{ipcw.3}}$
0	0	50	54.6	10.2	8.6	8.6	8.7	0.015	0.043	0.044	0.046
0	0.7	50	67.4	16.5	14.2	13.6	13.7	0.034	0.024	0.061	0.062
0	1.3	50	75.3	31.3	28	32.9	33.1	0.063	0.003	0.494	0.505
0.7	0	58	48.1	8	7.2	7.2	7.2	0.15	0.033	0.014	0.005
0.7	0.7	58	61.2	12.4	11.6	9.8	9.8	0.322	0.478	0.002	0.005
0.7	1.3	57.8	71.2	24.2	23.7	18	18	0.535	0.613	0.172	0.177
1.3	0	63.9	43	6.6	6.2	6.1	6.1	0.313	0.012	0.008	0.019
1.3	0.7	63.7	55.5	8.9	9.1	7.2	7.1	0.499	0.891	0.004	0.012
1.3	1.3	63.3	66.6	17.4	18.9	10.4	10.4	1.422	1.781	0.022	0.016
2	0	69.1	38	5.3	5	4.9	4.9	0.479	0.002	0.021	0.03
2	0.7	68.8	49	6.6	7.8	5.5	5.5	0.579	1.234	0.033	0.036
2	1.3	68.2	60.3	12.9	16.9	6.6	6.5	1.973	2.724	0.058	0.053

The column Cens shows average percentage of censored over 10,000 runs. The table shows the mean squared error (MSE) and the absolute bias. For the IPCW estimates, the weights were estimated with the marginal Kaplan–Meier estimate for the censored times $\left(\hat{\mathcal{C}}_{\text{ipcw.1}}\right)$, with a Cox estimate for the censored times $\left(\hat{\mathcal{C}}_{\text{ipcw.2}}\right)$, and with the stratified Kaplan–Meier estimate for the censored times $\left(\hat{\mathcal{C}}_{\text{ipcw.3}}\right)$, respectively.

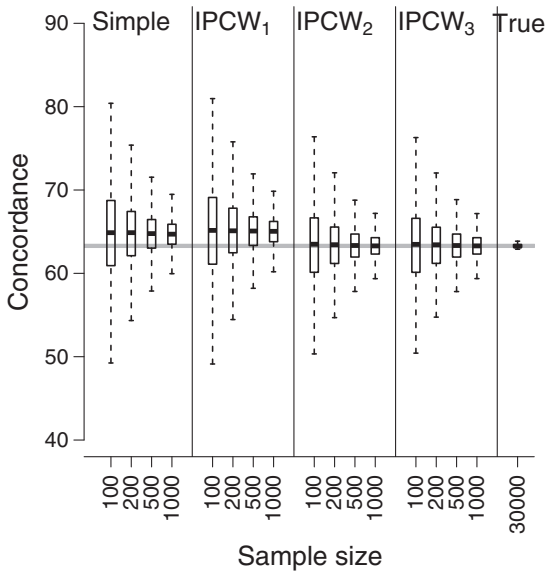


Figure 1. Results of the simulation study with a single binary predictor. Boxplots of results of 10,000 simulation runs are shown for $\beta = \gamma = 1.3$ and varying sample sizes.

modeling approaches were tried and compared by means of Somers' rank correlation coefficient. This is defined as $\mathcal{D} = 2 * \tilde{\mathcal{C}}(t) - 1$ where $\tilde{\mathcal{C}}(t)$ is the unrestricted version of the concordance index for prediction horizon t (Section 2.3).

4.1. The data

The primary endpoint used in the study is time to tumor recurrence which was observed for 268 of the 1042 enrolled patients. The remaining 774 patients were right censored at the end of their follow-up time. There are five predictors considered: prostate-specific antigen (PSA), clinical stage (CS), Gleason score sum (GSS), radiation dose (RD), and hormone therapy (HT). The predictors PSA, GSS, and RD

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are on a continuous scale; HT is binary, either yes or no; and CS is categorical, taking one of the values (T1c, T2a, T2b ,T2c, T3ab, and T3c).

4.2. *Prediction models*

The original study [12] compared eight prediction models with Somers' rank correlation coefficient, which is a simple transformation of the concordance index. Here we selected two of the previously considered prediction models. The first model uses the following linear predictor:

$$\text{Model 1 (Cox): } \log(PSA) + CS + HT + GSS + RD), \tag{7}$$

and the second uses restricted cubic splines to estimate the functional form of the relation between the continuous predictors and the log-hazard rate:

$$\text{Model 2 (CoxSpline): } \text{rcs}(\log(PSA), 3) + CS + HT + \text{rcs}(GSS, 3) + \text{rcs}(RD, 3).$$

Both models are fitted using the statistical software R with the routine *cph* of the R package *rms* [18]. Results for Model 1 are shown in Table II. For Model 2, $\text{rcs}(x, 3)$ denotes a restricted cubic spline with three knots applied to predictor x.

For the purpose of illustration, we also consider two prediction models not considered previously. Model 3 is derived from Aalen's nonparametric additive hazard regression model [14] which allows that predicted survival probabilities of two patients can cross in time. Model 3 is fitted with the function *aalen* of the R package *timereg* and uses the same linear predictor as the first Cox regression model. On the basis of the data of the prostate cancer patients, the Aalen regression model identifies a significant time-varying cumulative regression coefficient of hormone therapy. The effect is first positive and then turns sign showing a negative after 20 months. Model 4 was a random survival forest model based on 1000 survival trees [15] fitted with the R package *randomSurvivalForest*. Model 4 differs from the other models as it belongs to the machine learning culture [19].

4.3. *Working model for censoring weights*

By study design, the dose of radiation therapy depends on when the patient entered the study, and there were strong effects of almost all predictors on the censoring mechanism. Figure 2 shows the dependence of the censoring distribution on the RD. We estimated the weights $\widehat{W}_{ij}$ based on a working Cox regression model for the censoring hazard (Table II).

4.4. *Bootstrap cross-validation of concordance index*

For the four models, we estimated the truncated concordance index for prediction horizons on a grid of time points between the time origin and 6 years of length 6 months. At each prediction horizon t and for each model, we estimated the concordance index on the basis of (the same) 1000 bootstrap samples $D_1, \ldots, D_{1000}$ drawn with replacement from the original data. In each bootstrap, we repeated all steps

Table II. Results from Cox regression applied to the survival times and censoring times.

	Survival distribution			Censoring distribution		
Predictor	Hazard ratio	CI 95	*p*-value	Hazard ratio	CI 95	*p*-value
Prostate-specific antigen	1.003	[1.001; 1.004]	0.0043	0.994	[0.989; 0.999]	0.0172
Radiation dose	0.630	[0.544; 0.731]	< 0.0001	2.389	[2.176; 2.622]	< 0.0001
Gleason score sum	1.197	[1.095; 1.308]	< 0.0001	1.032	[0.978; 1.089]	0.2530
Clinical stage = T1c	1	—	—	1	—	—
Clinical stage = T2a	1.662	[0.984; 2.806]	0.0573	1.074	[0.858; 1.345]	0.5345
Clinical stage = T2b	2.446	[1.580; 3.787]	< 0.0001	0.701	[0.569; 0.863]	0.0008
Clinical stage = T2c	3.453	[2.235; 5.335]	< 0.0001	0.404	[0.320; 0.509]	< 0.0001
Clinical stage = T3ab	3.039	[1.810; 5.102]	< 0.0001	0.387	[0.289; 0.519]	< 0.0001
Clinical stage = T3c	4.183	[2.704; 6.468]	< 0.0001	0.372	[0.285; 0.484]	< 0.0001
Hormone therapy = yes	0.915	[0.689; 1.215]	0.5402	1.443	[1.240; 1.679]	< 0.0001

The radiation dose values have been centered at the mean value and scaled with root mean square value.

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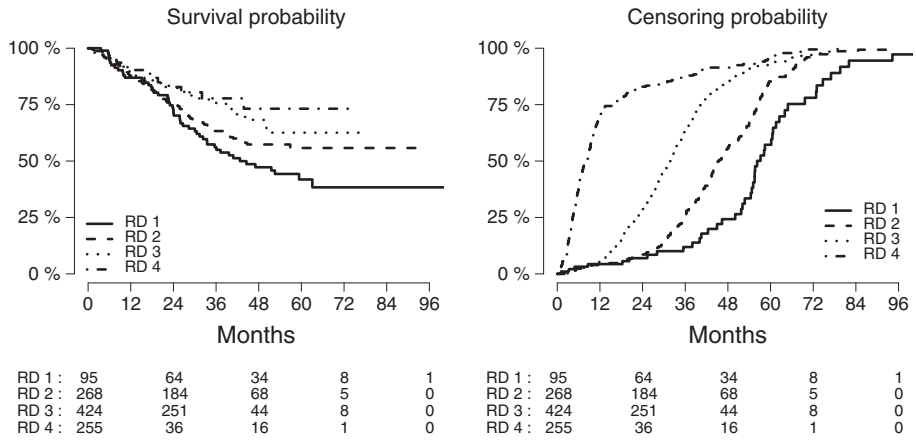


Figure 2. The survival and the censoring probability in four radiation dose strata (RD1: [6000,7000), RD2: [7000,7500), RD3: [7500,8000), RD4: [8000,8640]). In the left panel, the curves are estimated with the usual Kaplan–Meier estimator. In the right panel, the Kaplan–Meier estimator is used for the censoring times. The numbers at risk are given below the figure.

of modeling: for Models 1 and 3 the regression coefficients and baseline hazards were estimated; for Model 2 the cubic splines; and for Model 4 new survival trees were built in each bootstrap sample. Using the resulting prediction models, we predicted the outcome for all patients *not drawn* in the current bootstrap sample. For each bootstrap b, this yielded risk orders $1\{M_b(t, X_i) > M_b(t, X_j)\}$ for all pairs $i, j \notin D_b$ where M_b is the model fitted to the bth bootstrap set. We estimated the concordance index as the average over bootstrap samples using the following formula:

$$\hat{\mathcal{C}}_{\text{ipcw},B}(t) = \frac{1}{1000}\sum_{b=1}^{1000}\frac{\sum_{i\notin\mathcal{D}^b}\sum_{j\notin\mathcal{D}^b} 1\{M_b(t,X_i) > M_b(t,X_j)\}1\left\{\widetilde{T}_i < \widetilde{T}_j\right\}\widetilde{N}_i(t)\widehat{W}_{ij}}{\sum_{i\notin\mathcal{D}^b}\sum_{j\notin\mathcal{D}^b} 1\left\{\widetilde{T}_i < \widetilde{T}_j\right\}\widetilde{N}_i(t)\widehat{W}_{ij}}. \quad (8)$$

We derived the weights $\widehat{W}_{ij}$ only once using the full data set.

4.5. *Results*

Figure 3 shows the estimated concordance index of the four models over time. At the 1-year prediction horizon, the Aalen model (Model 3) wins with $\hat{\mathcal{C}}_{\text{ipcw},B}(12) = 77.47\%$, followed by the random survival forest (Model 4) $\hat{\mathcal{C}}_{\text{ipcw},B}(12) = 76.2\%$, the spline version of the Cox model (Model 2) $\hat{\mathcal{C}}_{\text{ipcw},B}(12) = 75.6\%$, and the Cox model (Model 1) $\hat{\mathcal{C}}_{\text{ipcw},B}(12) = 74.67\%$. At the 5-year prediction

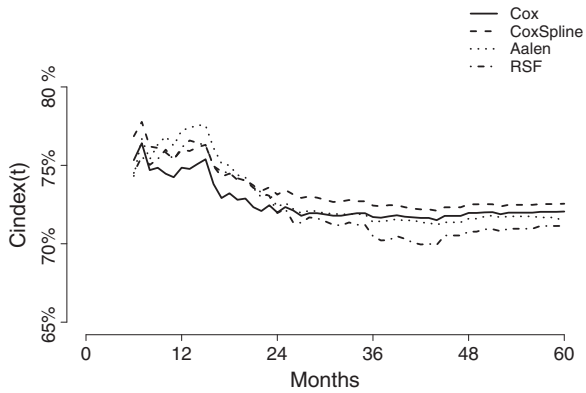


Figure 3. Discrimination curves for the four models estimated using the bootstrap cross-validation procedure described in Section 4.4. The models predict the event risks at the same time points (x-axis) where the concordance index is truncated.

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horizon, the picture has changed: the spline Cox model wins with $\hat{C}_{\text{ipcw},B}(60) = 72.8\%$, followed by the Cox model (Model 1) with $\hat{C}_{\text{ipcw},B}(60) = 72.3\%$, the Aalen model (Model 3) $\hat{C}_{\text{ipcw},B}(60) = 71.8\%$, and the random survival forest (Model 4) $\hat{C}_{\text{ipcw},B}(60) = 71.4\%$.

The corresponding bootstrap cross-validation results, obtained with the simple statistic in the same bootstrap samples, at 60 months, also yielded $\hat{C}_{\text{simp},B}(60) = 71.24\%$ for Model 1, $\hat{C}_{\text{simp},B}(60) = 72.16\%$ for Model 2, and $\hat{C}_{\text{simp},B}(60) = 70.64\%$ for Model 3. Regardless, all differences were small.

4.6. Sensitivity analysis

We simulated data on the basis of the prostate cancer data set to study the sensitivity of the results with respect to misspecification of the working model for the censoring distribution. To generate data with a similar distribution as the observed data, we first fitted two parametric Cox–Weibull regression models [17]: one for the event time hazard and another for the censoring time hazard. In order to avoid extremely censored data sets, the censoring baseline hazard was reduced by a factor of 10% so that the simulated data had only 50% censored event times on average. Then the joint distribution of the predictor variables was approximated by normal distributions for log(*PSA*), *GSS*, and *RD* using observed means and variances and by binomial distributions for *HT* and *CS* using observed frequencies. The result of both steps yielded a fully parametric model for the joint distribution of (T_i, C_i, X_i). The main simplification was that we assumed independence between the predictor variables.

From this joint model, we generated 10,000 data sets of size 500. We predicted the outcome in each data set with the Cox regression model (Model 1, equation (7)), which was estimated with the original prostate cancer data. In each data set, the truncated concordance index at $t = 36$ months was estimated using IPCW with four different models for the conditional censoring distribution: (1) a Cox regression model with the same linear predictor used for generating the data; (2) a Cox regression model where the predictor variable *RD* was dropped from the linear predictor; (3) a Cox regression model where the

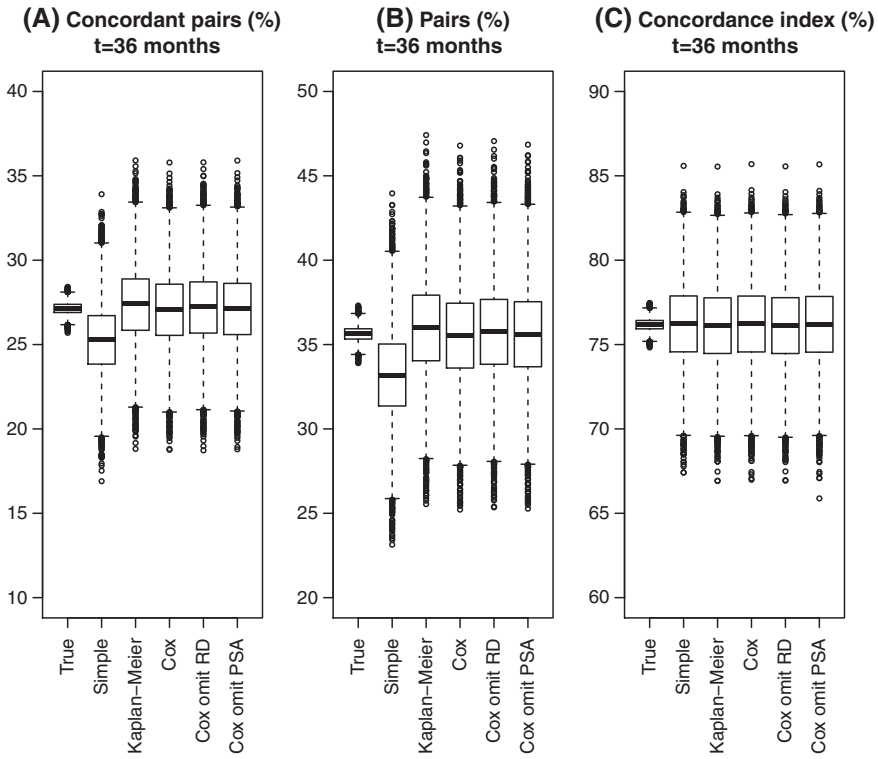


Figure 4. Boxplots showing results from the sensitivity analysis. The concordance index of the Cox model fitted in the original data was estimated with four different IPCW estimates and the simple statistic in 10,000 simulated data sets each of size 500. The true value was approximated based on 10,000 simulated uncensored data sets of size 20,000. Panels (A) and (B) show estimates of the numerator and denominator of the concordance index defined as $\hat{A}$ and $\hat{B}$ in equations 6 and 3, respectively. Panel (C) shows the estimated concordance index.

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variable PSA was dropped from the linear predictor; (4) assumed independence between the censoring and all predictor variables using the marginal Kaplan–Meier estimate for the censoring distribution. In addition, the truncated version of the simple statistic (3) was computed. The true concordance index was estimated as the average concordance index calculated in 10,000 simulated data sets of sample size 20,000 and using the uncensored event times.

The results are shown in Figure 4. Interestingly all estimates and also the simple statistic yield comparable results for estimating the truncated concordance index. Only when the numerator, $\hat{A}$, and the denominator, $\hat{B}$, of the concordance index were considered separately, we observed a large sample bias for the simple statistic (equation (6)) and for the IPCW statistic (equation (3)) based on misspecified censoring models.

5. Discussion

In the medical literature, the concordance index was typically evaluated at the maximum follow-up time in the study and by ignoring unusable pairs. However, in most applications it would be preferable to truncate the concordance index at an earlier time point than the maximum follow-up time. This is for the same reason that one should not interpret the Kaplan–Meier estimate at the maximum follow-up time where the number at risk is small. As we have illustrated with the prostate cancer data, it may be useful to evaluate the truncated concordance index at several time points and to follow the discrimination ability over time.

By simply ignoring unusable pairs, results become dependent on the follow-up period (censoring distribution) which can be controlled by the study design. This problem can be solved by estimating suitable weights. In this article, we have proposed a class of estimates that rely on a working model for the conditional probabilities of being uncensored. The advantage of the proposed IPCW estimate is that it does not assume anything directly about the relation between survival times and predictor variables. The method can thus be used to assess prediction models of any kind, for example, Aalen regression or random survival forest, and is not restricted to Cox regression models. The disadvantage of the IPCW estimate is that one has to model the censoring distribution. If it is reasonable to assume that the censoring times are independent of the predictors then the probability of being uncensored can be estimated by the Kaplan–Meier estimator for the censoring times [8, 9]. This assumption is clearly violated in the application considered in this article (Section 4). In the prostate cancer study, the patient individual follow-up time depended strongly on the predictor variables. Our simulation results (Section 3.3) indicated that a misspecified working model can introduce a bias in the IPCW estimate which is of a similar magnitude as the bias of a truncated version of the simple statistic. In this situation, a regression model for the distribution of the censoring times avoids the bias. Our sensitivity study (Section 4.6) illustrated that for the prostate cancer data, the IPCW estimate was robust against misspecification of the working model for the censoring distribution. Surprisingly, it also showed that the truncated version of the simple statistic, which simply ignores 'unusable' pairs, performed reasonably good, as the bias incurred in the numerator was leveling out the bias incurred in the denominator.

5.1. Other assessment measures

The concordance index as considered here uses the survival time as response. It thus differs conceptually from the generalization of the area under the ROC curve for survival analysis [20, 21] which uses the binary survival status at given time points as response. The latter quantifies how well the prediction model can order the survival status at a given time point; the former quantifies how well the prediction model can order the survival times. Heagerty and Zheng established the formal relation between the two measures [7]. Because both the Area under the ROC curve and the concordance index are rank statistics, they should be accompanied by a measure of the calibration of the prediction model. At a single prediction horizon, a calibration curve can be used to compare observed and predicted outcomes. Another useful measure of calibration is the time-dependent Brier score [16, 22, 23].

5.2. Assessment of all modeling steps

Previous work on the concordance index was often restricted to pre-specified models such as Cox regression with a known linear predictor. However, in real applications, most prediction models result from applying flexible modeling strategies, including variable and functional form selection, or tools

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from machine learning. It is well known that naive calculation of the concordance index in the training data, which were used to derive the models, may give flawed results in favor of overfitting models. In the absence of independent validation data, natural remedies for this 'apparent error problem' are versions of cross-validation where all steps of the modeling are repeated in subsets of the data (or in bootstrap samples) [24, 25]. In the present article, we have used a cross-validation estimate of the concordance index based on bootstrap samples of the prostate cancer data. More advanced estimates such as the optimism-corrected bootstrap [26] or the .632+ estimate [24] may not have worked here for the random survival forest model, which has an almost perfect apparent performance.

5.3. Software

The truncated version of the simple statistic and the IPCW estimates presented in this article can be estimated with the function *cindex* of the R package *pec* [27, 28] available at http://cran.r-project.org/. The function uses either independent validation data or cross-validation and optionally allows different treatments of ties (equal values) in event times and predicted values.

Appendix A. Asymptotic bias of the simple concordance statistic

To express the asymptotic bias of the truncated version of the simple statistic defined in (3), first note that under conditional independence one has

$$E_i\left(\widetilde{N}_i(t) \mid X_i\right) = -\int_0^t G(u- \mid X_i)S(\mathrm{d}u \mid X_i).$$

By also using the independence of subjects i and j and equation (5), this yields

$$E_{ij}\left\{\widetilde{N}_i(t)1\left\{\widetilde{T}_i < \widetilde{T}_j\right\} \mid X_i, X_j\right\} = \int_0^t G(u- \mid X_i)G(u \mid X_j)S(u \mid X_j)S(\mathrm{d}u \mid X_i).$$

One shows that for $m \to \infty$ and fixed training set D_n^{train}, the simple concordance statistic $\hat{\mathcal{C}}_{\text{simp}}(t)$ converges in probability to

$$\mathcal{C}^*_{\text{simp}}(t) = \frac{E_{ij}\left\{1\{M_n(t,X_i) > M_n(t,X_j)\}\int_0^t G(u- \mid X_i)G(u \mid X_j)S(u \mid X_j)S(\mathrm{d}u \mid X_i)|D_n^{\text{train}}\right\}}{E_{ij}\int_0^t G(u- \mid X_i)G(u \mid X_j)S(u \mid X_j)S(\mathrm{d}u \mid X_i)}.$$

Subtracting the true value (4) shows that the asymptotic bias $\mathcal{C}^*_{\text{simp}}(t) - \mathcal{C}(t)$ has the form

$$\frac{E_{ij}\left\{1\{M_n(t,X_i) > M_n(t,X_j)\}\int_0^t\left[P(T_i<T_j)G(u- \mid X_i)G(u \mid X_j) - P\left(\widetilde{T}_i < \widetilde{T}_j\right)\right]S(u \mid X_j)S(\mathrm{d}u \mid X_i)|D_n^{\text{train}}\right\}}{P(T_i < T_j)P\left(\widetilde{T}_i < \widetilde{T}_j\right)}.$$

Appendix B. Consistency and bias of the IPCW estimates

Under (A1)–(A3), $\widehat{G}_m(t- \mid X_i)\widehat{G}_m(t \mid X_j)$ consistently estimates the weights $G(t- \mid X_i)G(t \mid X_j)$ and it follows from the same arguments used in Section A that for $m \to \infty$ and fixed n, the IPCW estimate $\hat{\mathcal{C}}_{\text{ipcw}}(t)$ converges in probability to

$$\frac{E_{ij}\left\{1\{M_n(t,X_i)>M_n(t,X_j)\}\int_0^t\{G(u-|X_i)G(u|X_j)\}^{-1}G(u-|X_i)G(u \mid X_j)S(u \mid X_j)S(\mathrm{d}u \mid X_i)|D_n^{\text{train}}\right\}}{E_{ij}\int_0^t\{G(u-|X_i)G(u|X_j)\}^{-1}G(u-|X_i)G(u \mid X_j)S(u \mid X_j)S(\mathrm{d}u \mid X_i)} = \mathcal{C}(t).$$

In practice, a bias is introduced if (A3) is violated and the model for G misspecified. Suppose $\widehat{G}_m$ converges in probability to $G^* \neq G$. Then, as $m \to \infty$ the weights W_{ij} converge to $W^*_{ij}(t) = G^*(t \mid X_i)G^*(t- \mid X_j)$ and $\hat{\mathcal{C}}_{\text{ipcw}}(t)$ converges to

$$\mathcal{C}^*_{\text{ipcw}}(t) = \frac{E_{ij}\left\{1\{M_n(t,X_i) > M_n(t,X_j)\}\int_0^t\{W^*_{ij}(u)\}^{-1}G(u- \mid X_i)G(u \mid X_j)S(u \mid X_j)S(\mathrm{d}u \mid X_i)|D_n^{\text{train}}\right\}}{E_{ij}\left\{\int_0^t\{W^*_{ij}(u)\}^{-1}G(u- \mid X_i)G(u \mid X_j)S(u \mid X_j)S(\mathrm{d}u \mid X_i)\right\}}.$$

Then $\mathcal{C}^*_{\text{ipcw}}(t) - \mathcal{C}(t)$ is the asymptotic bias of $\hat{\mathcal{C}}_{\text{ipcw}}(t)$.

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